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Introduction to Ophthalmic Disorders

Alanna Nattis, DO FAAO
Clinical Professor of Ophthalmology and Surgery NYIT-COM
Cornea, Cataract and Refractive Surgeon
Director of Clinical Research, SightMD
ANATTIS@nyit.edu

Ocular Anatomy: Cornea

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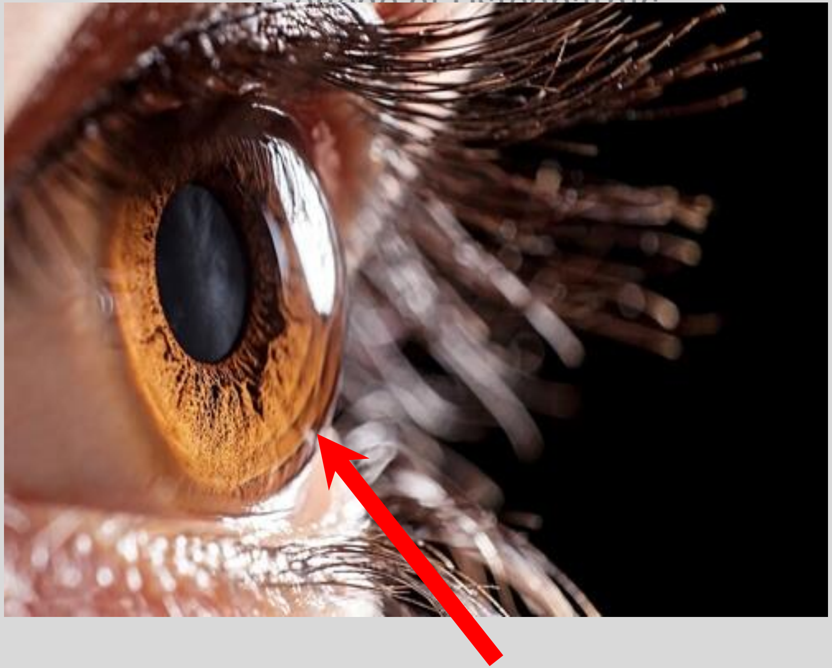
- Transparent anterior portion of the eyeball
  - Protects intraocular contents
  - 75% of refractive power of the eye
    - LASIK
  - Avascular: receives nutrition from tear film and aqueous humor
  - Rich sensory innervation: 300-400x that of skin
  - **5 layers**:
    1. Epithelium
    2. Bowmans' layer
    3. Stroma
    4. Descemet's membrane
    5. Endothelium



Cornea

Ocular Anatomy: Conjunctiva

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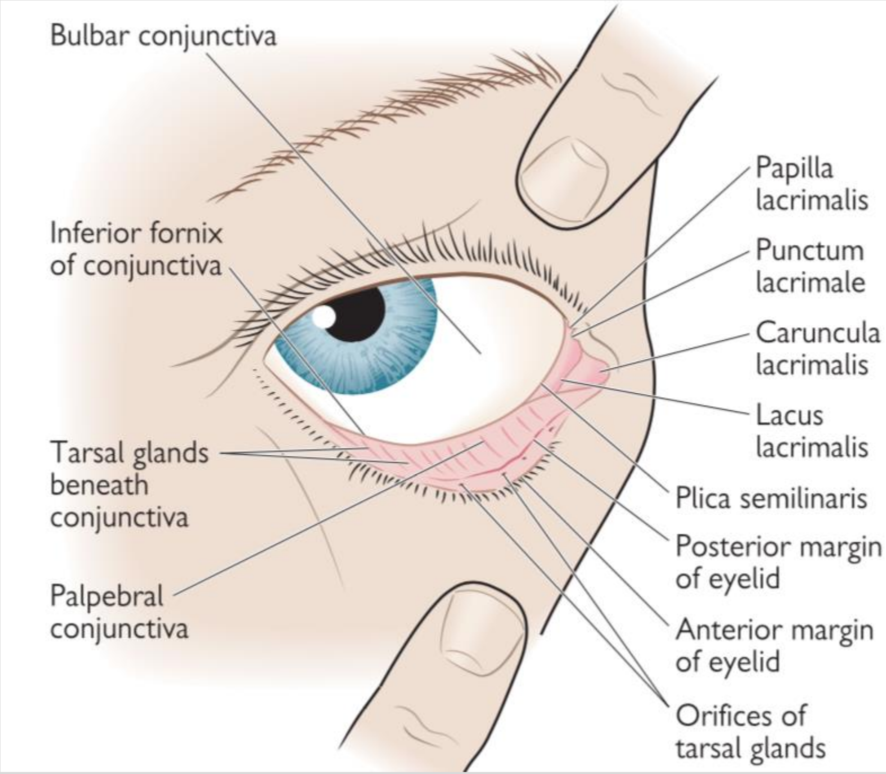
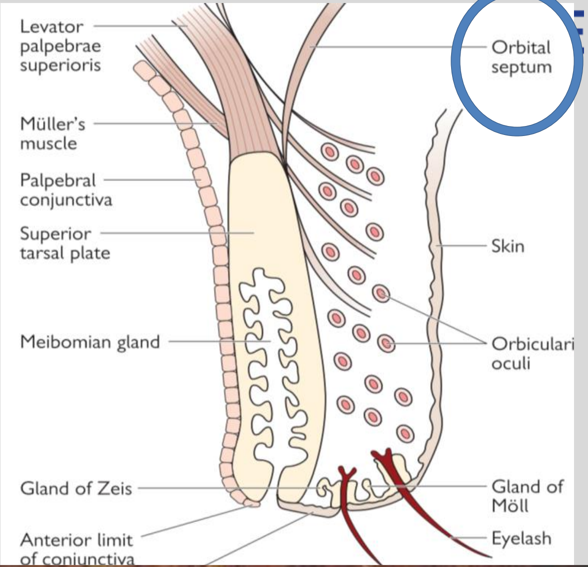
- Thin translucent mucous membrane
  - Lines eyelids (*palpebral conjunctiva*) and surface of globe (*bulbar conjunctiva*)
  - Goblet cells: contribute to mucin layer of tear film
  - Accessory lacrimal glands: create aqueous tears



Ocular Anatomy: Eyelids

- Protect ocular surface from mechanical injury and when closed, protects the retina from bright light
  - Blinking assists distribution of tears across the ocular surface
  - Gross structure: lids meet at lateral and medial canthus
  - **Meibomian glands** (~50 per eye): produce sebaceous secretion that constitutes outer lipid layer of tear film
  - **Glands of Moll**: modified **sweat** glands that open on to the lid margin between the lashes or into a lash follicle
  - **Glands of Zeis**: **sebaceous** glands that open into each follicle



Ocular Anatomy: Nasolacrimal system

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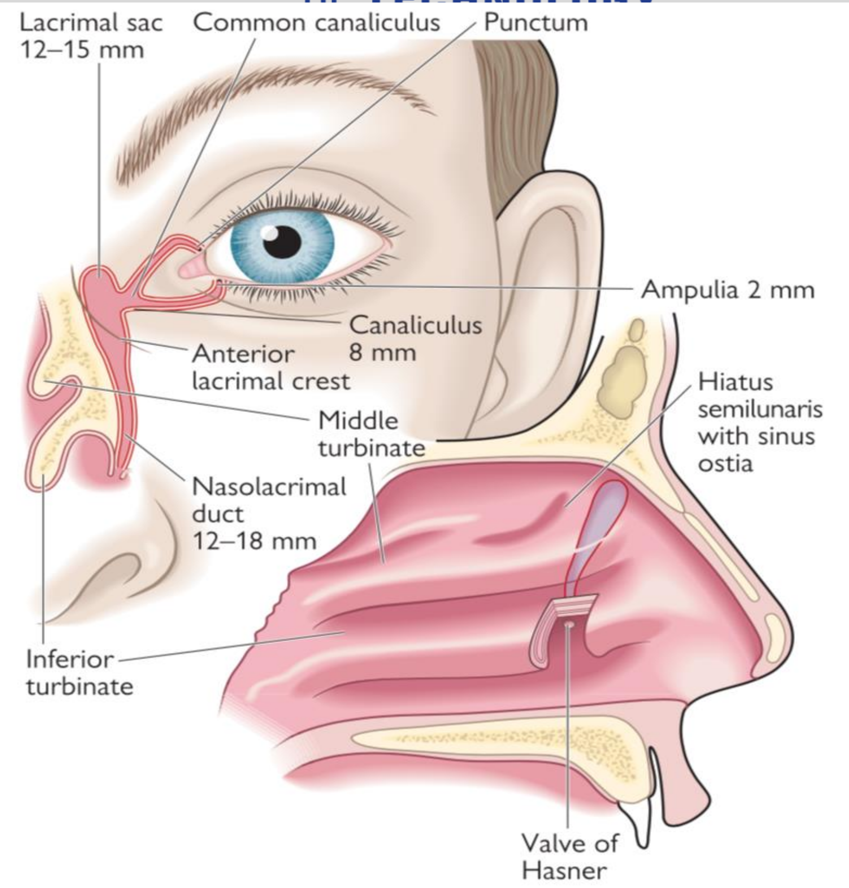
- **Tear production**:
  - main lacrimal gland
  - accessory lacrimal glands (in the conjunctiva)
- **Nasolacrimal drainage system**: conduit for the flow of tears from the external surface of the eye → nasal cavity
- **Lacrimal puncta**: small openings that lie on the medial aspect of the upper and lower lids
- **Lacrimal canaliculi**: begin at puncta superior and inferior and join together at common canaliculus to drain into nasolacrimal sac
- **Lacrimal sac**: 12mm long, within the lacrimal fossa in anteromedial portion of orbital wall
- **Nasolacrimal duct**: connects lower portion of lacrimal sac with the nasal cavity and opens into the inferior meatus of the nose



Ocular Anatomy: Posterior Segment/Fundus

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- **Retina**: 200um neural tissue that lines the posterior inner aspect of the eye
  - Contains photoreceptor cells (rods/cones)
  - **Macula**: oval retinal area containing yellow xanthophyll pigments and 2+ layers of ganglion cells
  - **Fovea**: Central part of macula
- **Vitreous**: located between lens and retina
  - majority (80%, ~4ml) of globe volume
  - 99% water
  - 1% collagen (type II) fibrils and associated hyaluronan→impart gel like consistency
- **Choriocapillaris**: fenestrated capillary network fed by choroidal arterioles and drained by venules located in outer layer of choroidal stroma

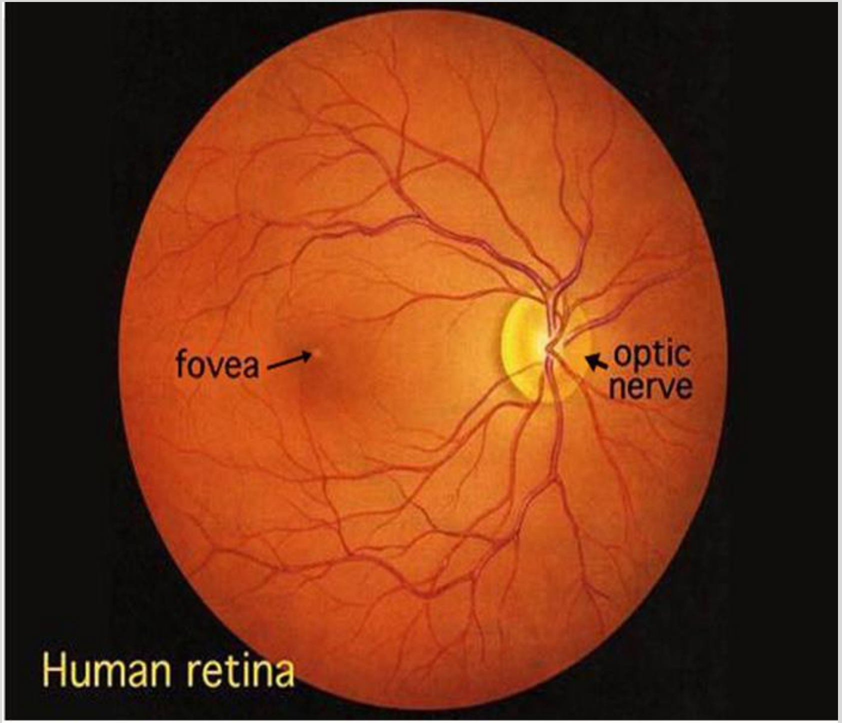
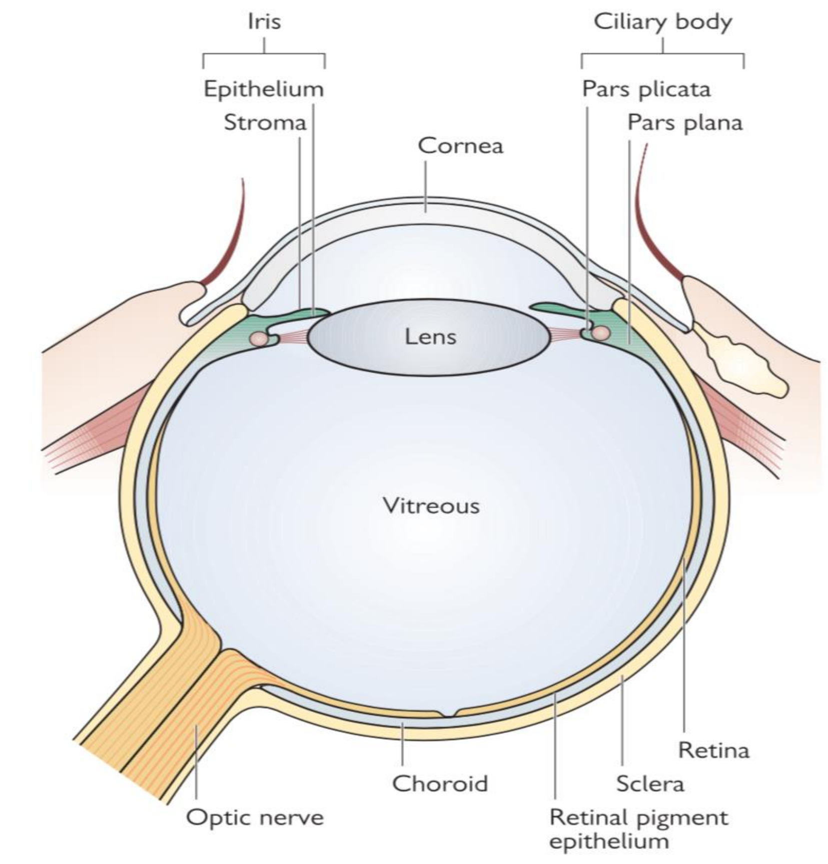


Figure 1. A view of the retina seen though an ophthalmoscope.

Ocular Anatomy: Uvea

- **Iris**: thin, contractile diaphragm with central aperture (pupil)
  - Anterior extension of ciliary body
  - Color dependent on melanocyte density
- **Ciliary body**: circumferential structure surrounding the lens and vitreous base
  - Responsible for aqueous fluid production and accommodation
- **Choroid**: thin, highly vascular layer lying between the retina internally and sclera externally



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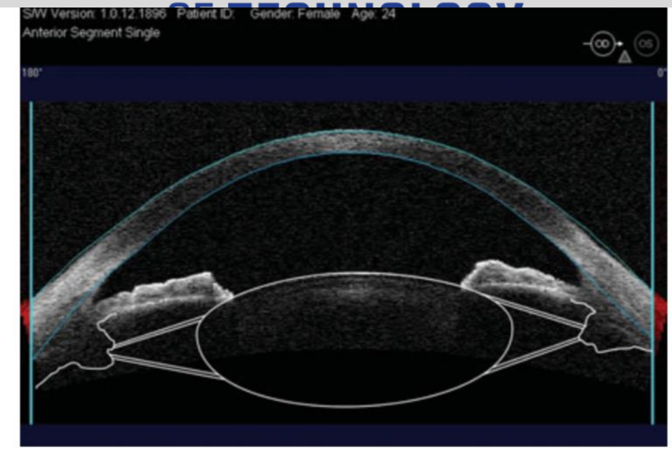
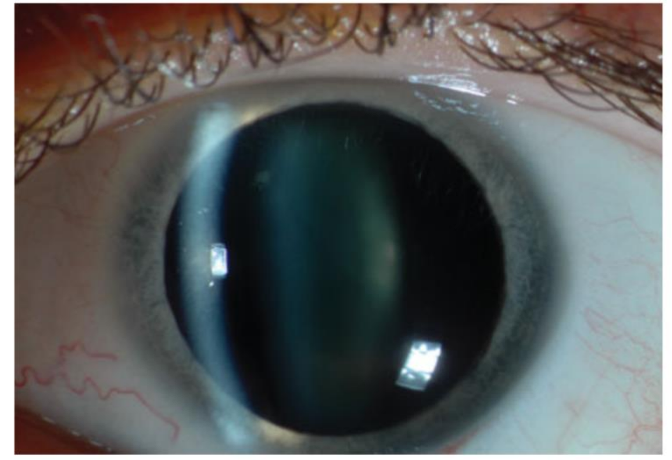
Ocular Anatomy: Lens

- Focuses images onto the retina, accounts for 1/3 refractive power of the eye
- **Zonules** support the lens within the eye: insert on lens equator circumferentially and are attached to ciliary body via ciliary processes
- **Accommodation**: ability of lens to change focus from distance to near; achieved by an increase in axial thickness of the lens and steepening of anterior and posterior radii of curvature
  - With age there is loss of accommodation: thus loss of clear near vision = need for reading glasses up close (***presbyopia***)





Ocular Anatomy: Cataract

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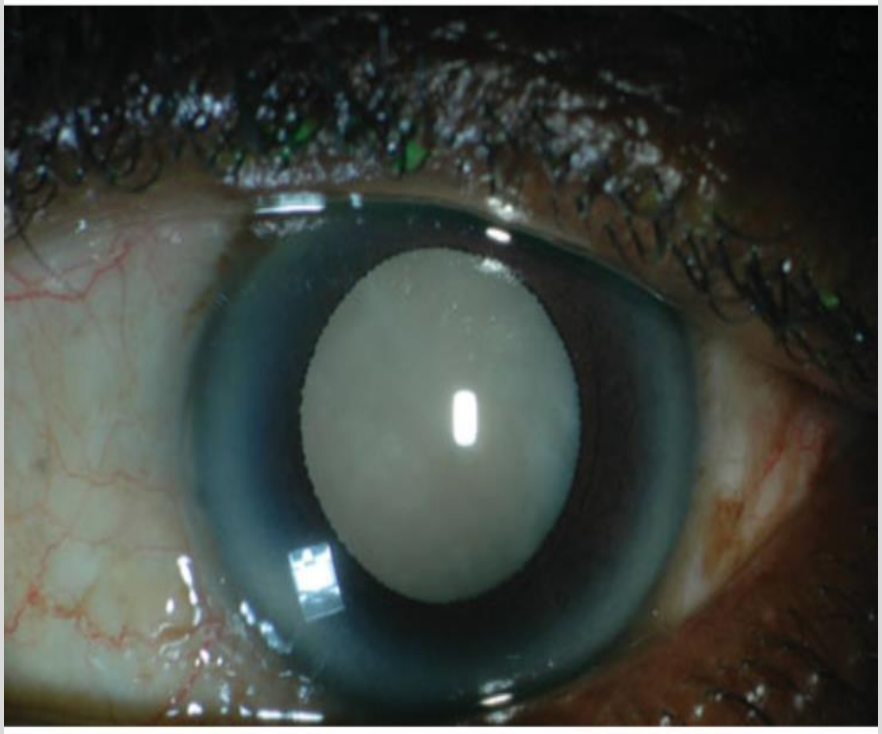
- Opacification of crystalline lens
- Age-related: MC
- *Drugs*: **corticosteroids**, chlorpromazine, amiodarone, topical glaucoma medications, miotic agents
- *Trauma*: penetrating/blunt injury, chemical injury, iatrogenic, electrocution, irradiation
- Secondary to *systemic disease*: **Diabetes**, myotonic dystrophy, Wilson's disease, atopic dermatitis, Neurofibromatosis, Fabry's disease
- Secondary to *ocular disease*: uveitis, myopia
- **Treatment**: cataract extraction with intraocular lens implantation
- Cataract Surgery



Ocular Anatomy: Sclera

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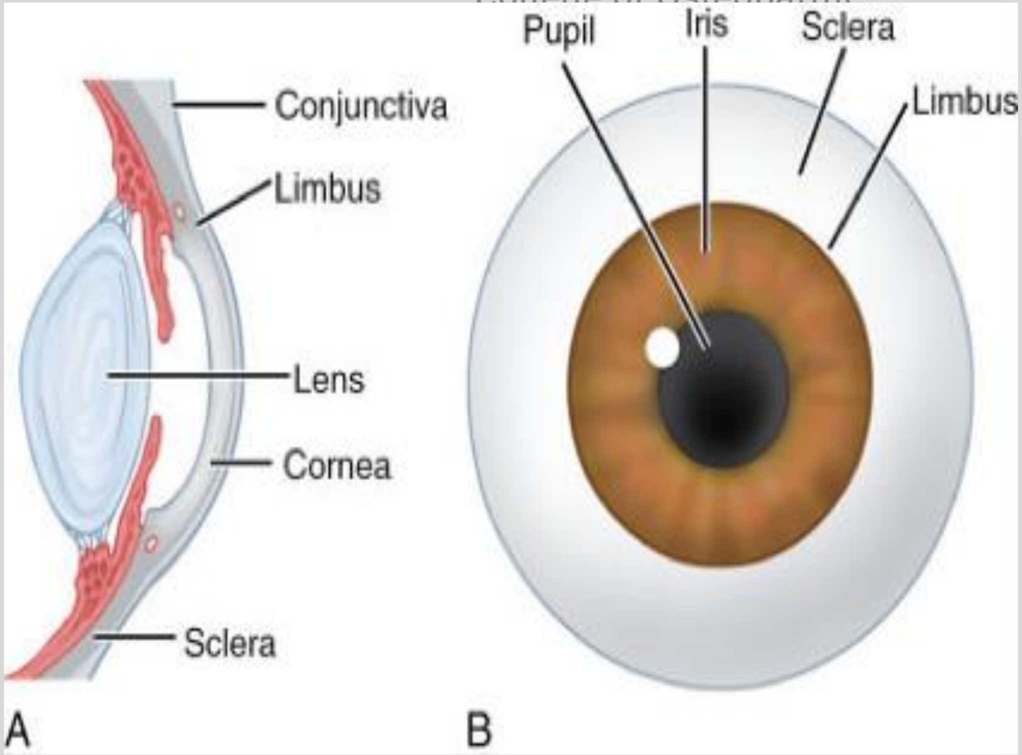
- Opaque, fibrous protective outer layer of the eye, white color
  - Thinner areas may appear blue
  - Continuous with dura mater and cornea
  - Maintains globe shape
  - Provides attachment for extraocular muscle (EOM) insertions



Ocular Anatomy: Extraocular muscles

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- move eyes in horizontal, *vertical* and **torsional** axes
- **Horizontal Muscles**:
  - Medial Rectus: adduction, Cranial nerve (CN)3
  - Lateral Rectus: abduction, **CN6**
- **Vertical Muscles**:
  - Superior Rectus: elevation, CN3
  - Inferior Rectus: depression, CN3
- **Oblique Muscles**:
  - Superior Oblique: intorsion, **CN4**
  - Inferior Oblique: extorsion, CN3

- ***Synergists***: muscles in the *same eye* that move the eye in the same direction
- ***Agonist-antagonist pairs***: muscles in ***same eye*** that have opposing actions
- ***Yoke muscles***: pairs of muscles in **opposite eyes** that move the 2 eyes in the same direction
- ***Strabismus***: improper alignment of the eyes

Basic Approach to "The Eye Patient"

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- HPI: OPQRST, visual complaints, diplopia (monocular, binocular)
  - Recent hx of trauma? Surgery? Illness?
  - Exposure to someone with infectious eye condition (e.g. conjunctivitis)?
- Visual Acuity (VA, with and without correction)
- Extraocular motility (EOM)
- Pupils: size, reactivity, consensual response?
- Intraocular pressure (IOP)
- External exam: periocular skin changes? Edema?
- Slit lamp exam
  - Eyelids, lashes, conjunctiva, sclera cornea, anterior chamber, iris, lens, vitreous, retina, retinal vasculature
- Ancillary testing: as needed

"The Eye Patient"

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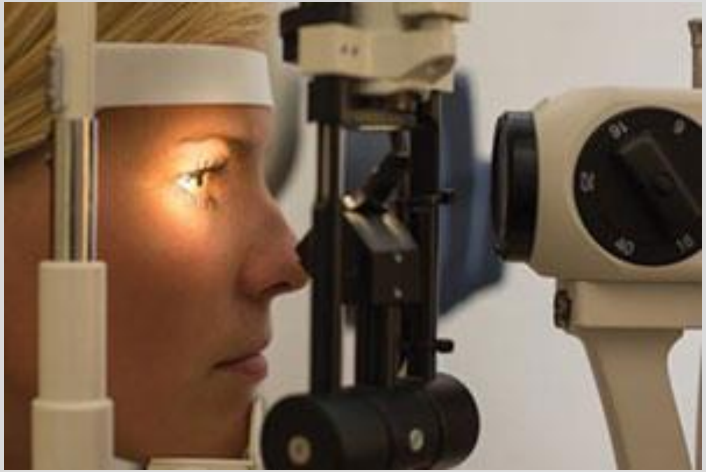
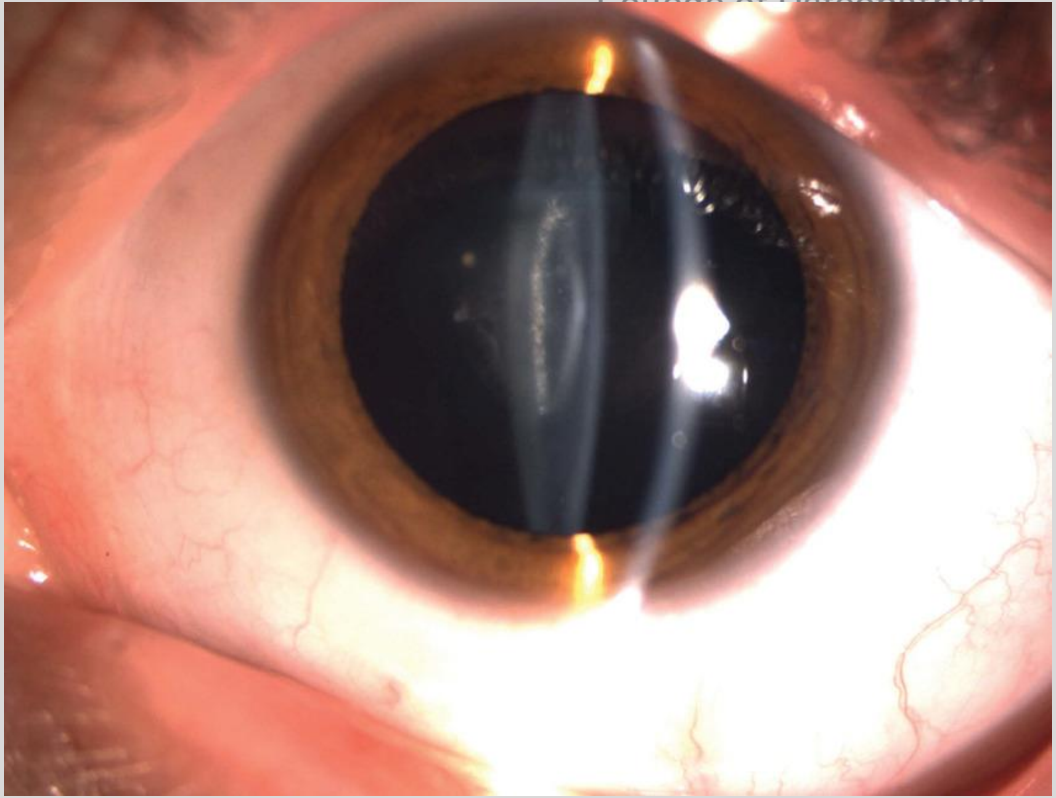
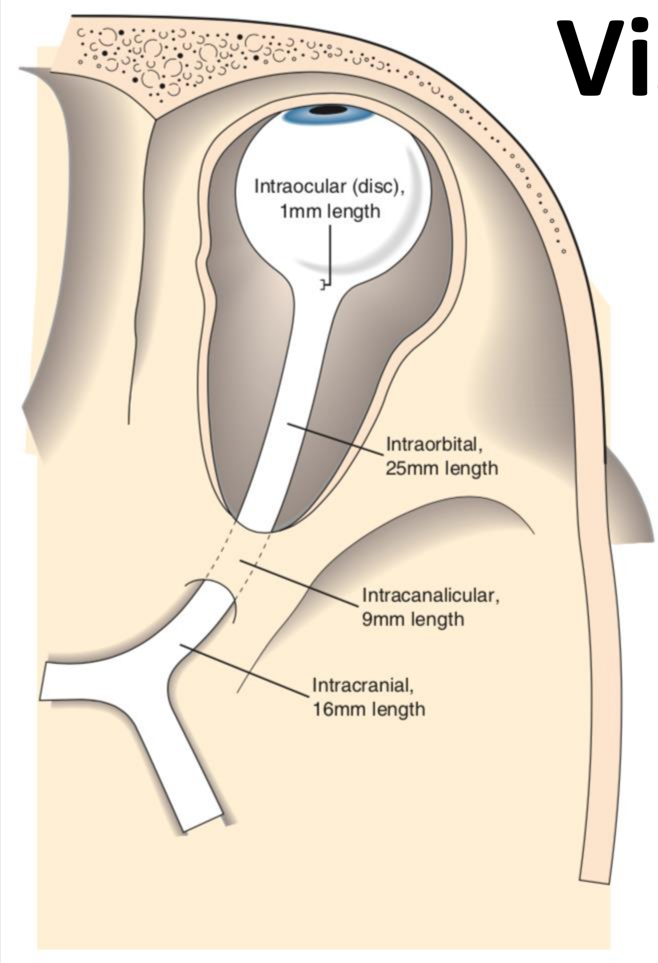


Slit Lamp



Visual Pathway



Optic Nerve

↓

Chiasm

↓

Optic Tract

↓

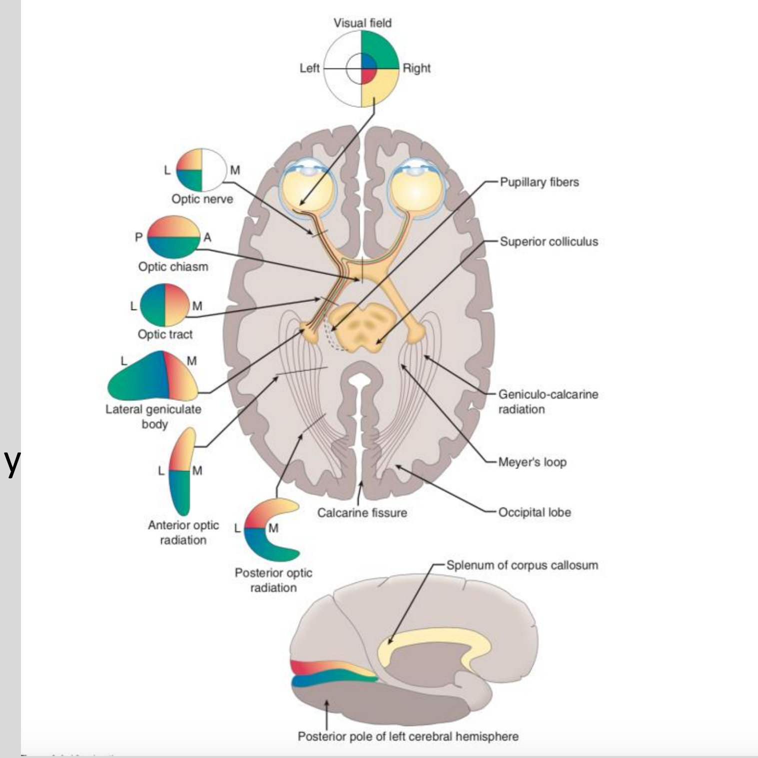
Lateral Geniculate Body

↓

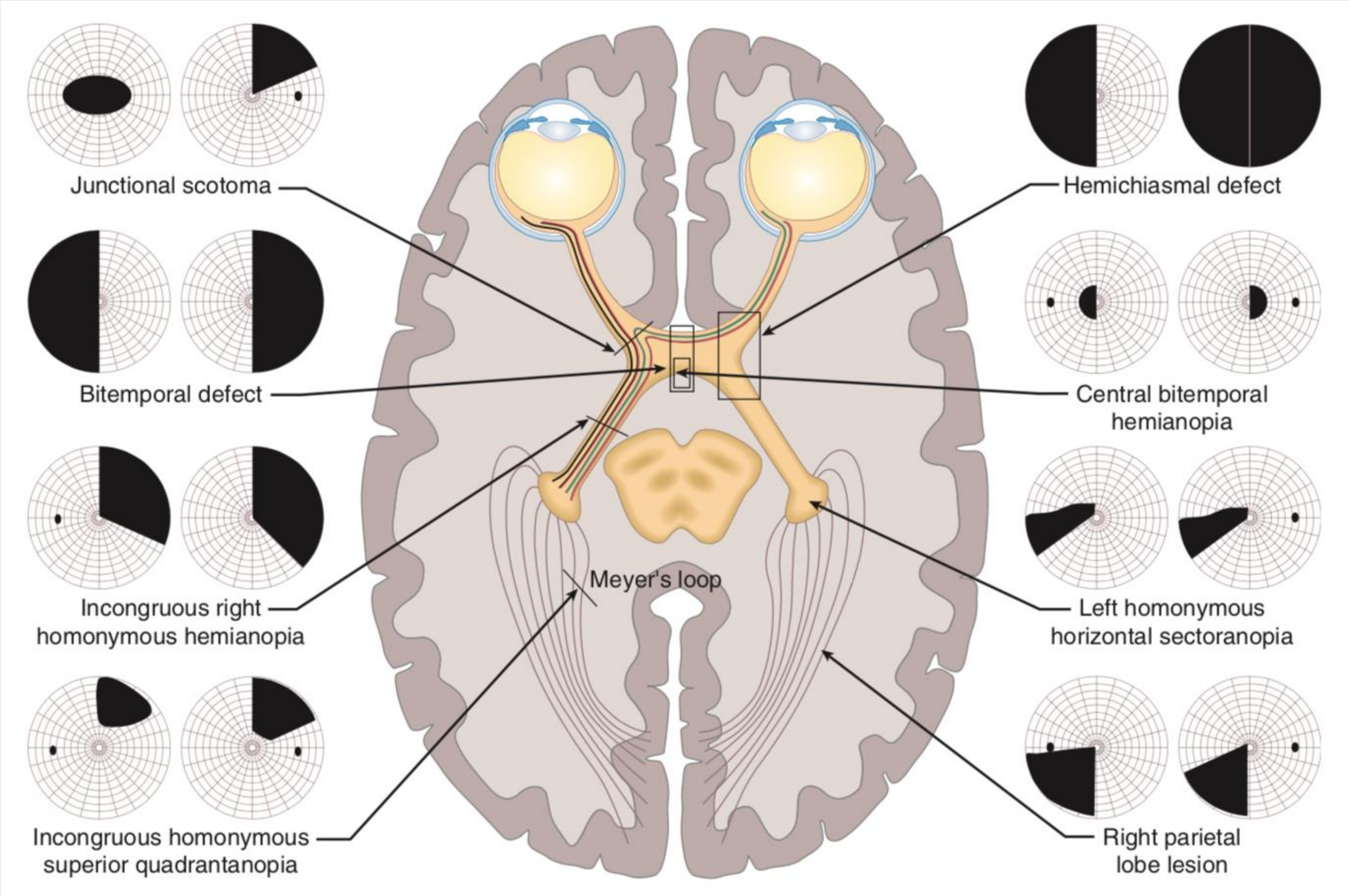
Optic Radiation

↓

Occipital Lobe



Visual Field Defects



Pupillomotor Pathway

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Light (image) enters Pupil

⇩

Optic Nerve

⇩

Chiasm (fibers split)

⇩

Optic Tract

⇩

Pretectal Nucleus (synapse)

⇩

Cross to both Edinger Westphal nuclei (synapse)

⇩

Travel via CN3 through subarachnoid space and cavernous sinus then travel **in** inferior division of CN3

⇩

Ciliary Ganglion

⇩

Postganglionic fibers travel via short ciliary nerves to Ciliary Body and Iris Sphincter

Pupils

- **Relative Afferent Pupillary Defect** (RAPD/Marcus-Gunn Pupil)
  - Light directed on one eye will *normally* induce consensual, constriction (miosis) in fellow eye. If there is an RAPD this does not occur (i.e. fellow eye will not react and/or dilate)
  - **Etiology**: large retinal lesion, asymmetric optic nerve disease (severe glaucoma), chiasm lesion, optic tract lesion (contralateral RAPD)
- **Light-Near Dissociation**: pupil does not react to light but near response is intact
  - **Etiology**: syphilis (Argyll-Robertson), Adie's pupil, familial dysautonomia (Riley Day syndrome), Parinaud's syndrome, encephalitis, alcoholism, herpes zoster ophthalmicus (HZO)
- **Anisocoria**: pupils of unequal size
  - ***Miosis***: small pupil
    - Etiology: Horner's syndrome, pharmacologic (brimonidine, narcotics, pilocarpine, insecticides), Argyll-Robertson pupil, iritis, diabetes
  - ***Mydriasis***: large pupil
    - Etiology: CN3 palsy, Adie's tonic pupil, pharmacologic (mydriatics, cycloplegics, cocaine), iris damage

Horner's Syndrome

- **"Ptosis, Miosis, Anhidrosis"**
  - Droopy upper lid
  - Small pupil
  - Lack of sweating on forehead above affected eye
  - Etiology:
    - **Preganglionic**: from hypothalamus to superior cervical ganglion
      - Vertebrobasilar stroke, vertebral artery dissection, cervical disc disease
    - **Postganglionic**: superior cervical ganglion to iris dilator
      - **Internal Carotid Artery Dissection**: TIA, stroke, neck pain, amaurosis, dyacusia, dysguesia
      - Cavernous sinus disease, Herpes Zoster, tumor of parotid gland/nasopharynx/sinuses

More Pupils

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- **Hutchinson's Pupil**: unilateral dilated, poorly reactive pupil in comatose patient due to ipsilateral supratentorial mass→displacement of hippocampal gyrus (*uncal herniation*) entrapping CN3
  – Pupillomotor fibers travel in peripheral portion of nerve and are susceptible to early damage from compression

Ocular Muscle Disorders

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- Chronic Progressive External Ophthalmoplegia (CPEO)
  - Mitochondrial disease
  - *Kearns-Sayre Syndrome*: ophthalmoplegia with ptosis, retinal pigment degeneration and heart block
- **Myasthenia Gravis**: weakness of voluntary muscles due to autoimmunity to motor end plates
  - antibodies block acetylcholine (Ach) Receptors (binding, blocking, modulating)
  - Fatigability is hallmark
  - *90% have **eye** symptoms*
    - 50% with eye symptoms: eye fatigue, diplopia, limited EOM's
    - 80% with ocular presentation will develop generalized disease
  - Ocular myasthenia: ptosis and weakness of EOM only
  - Systemic myasthenia: other skeletal muscles involved
    - 5% of patients develop Grave's disease
  - Dx: **Tensilon test** (Edrophonium IV) ← Acetylcholinesterase (AchE) Inhibitor that prolongs action of Ach→stronger muscle contraction
    - Positive 80-90% of the time, high false negative rate
    - May cause severe bradycardia (thus need to give test does of 1-2mg first and have atropine available for antidote if toxicity)
    - Rest test

Ocular Muscle Disorders

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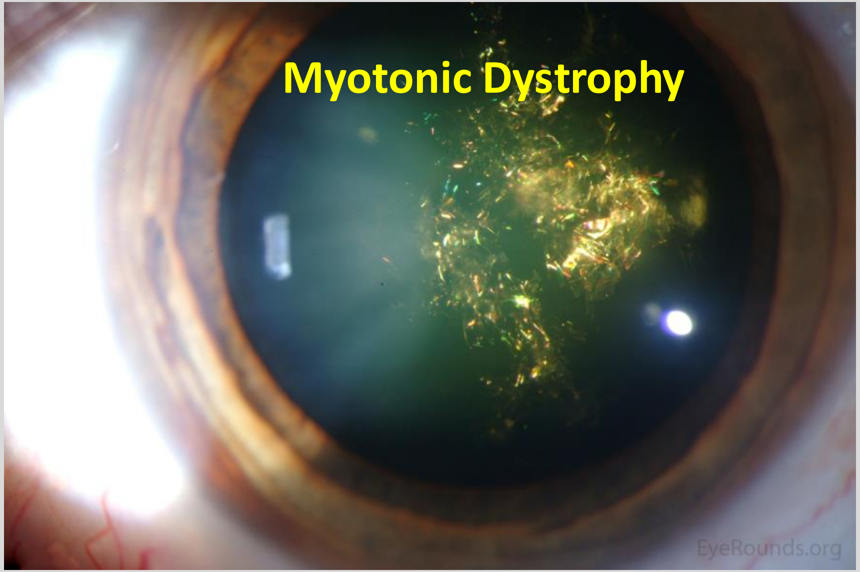
Thyroid related ophthalmopathy

- Dry eye, redness, pain
- Proptosis (GAG infiltration into EOM's)
- Restriction of motility
- Optic nerve compression
- Even if hormone levels are controlled, can still show stigma of thyroid eye disease (TED)



- **AD, Chr 19**
- Myotonia of peripheral muscles, worsens with cold, excitement, fatigue
- Christmas tree cataract, pigmentary retinopathy, ptosis, lid lag, light-near dissociation, miotic pupils, may develop very low intraocular pressure (hypotony)

Vascular Optic Neuropathies

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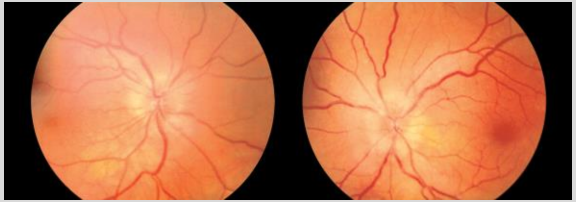
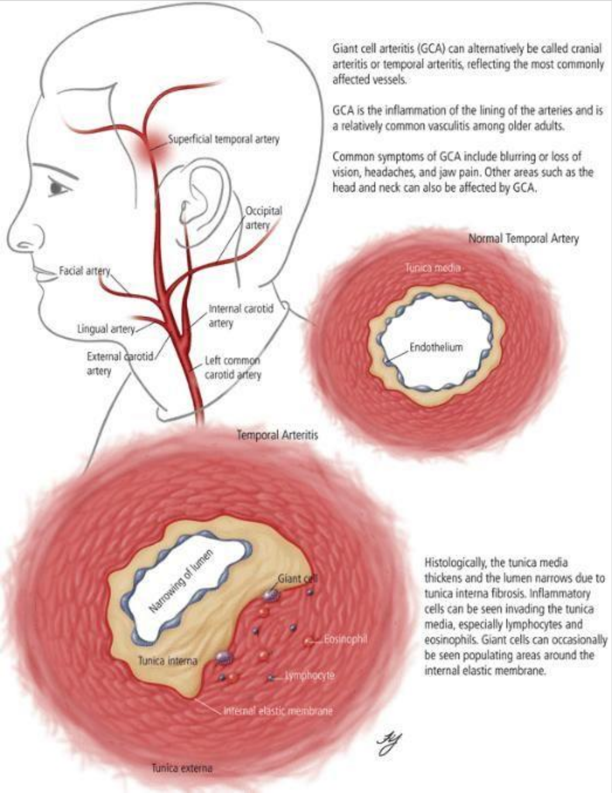
- **Anterior Ischemic Optic Neuropathy (AION):** infarct of optic nerve head just posterior to lamina cribrosa due to inadequate perfusion by posterior ciliary arteries→acute painless vision loss
  - Decreased color vision and visual acuity, RAPD, VF loss, unilateral optic disc edema, contralateral disc may appear small/crowded
  - **Arteritic ION (Giant Cell Arteritis/Temporal Arteritis)**: inflammatory vasculopathy affecting medium-large size vessels
  - **Nonarteritic ION (NAION)**: no associated sx, 50-75 y/o
    - Associated with DM, microvascular disease, collagen vascular disease, recurrence in same eye is rare, 25-40% fellow eye involvement, *NORMAL ESR*
- **Posterior ION**: acute, painless vision loss in one or both eyes
  - ↓ blood flow and oxygenation to the *intraorbital* optic nerve
  - ↓ in arterial perfusion pressure due to hypotension from volume blood loss
  - ↑ in peripheral vascular resistance causing reduction in downstream blood flow
  - ↑ in peripheral venous pressure from orbital edema
  - ↑ in intraocular pressure
  - ↓ in blood oxygen carrying capacity



Giant Cell Arteritis (Temporal Arteritis)

- Inflammatory vasculopathy affecting medium-large size vessels
  - F>M (2:1), age>55 y/o
  - +/- amarousis fugax/diplopia
  - **Systemic symptoms**: scalp tenderness, jaw claudication, joint pain (association with polymyalgia rheumatica), fever, malaise, anorexia, weight loss, anemia, Headache, tender temporal artery+/-decreased pulsation, neck pain, brain stem CVA
  - **Pathology**: granulomatous inflammation with epithelioid cells, lymphocytes, giant cells, disruption of internal elastic lamina, aneurysm formation
  - **Diagnosis**: elevated ESR, CRP, low hematocrit (HCT), thrombocytosis
  - **Treatment**: steroids (prednisone 60-100mg PO, +/- IV steroid initially) to prevent fellow eye involvement (65% risk of fellow eye involvement without treatment –usually affected within 10 days)



Idiopathic Intracranial Hypertension (*Pseudotumor*)

- Papilledema with NORMAL neuroimaging and CSF
- 90% female, mean age 33, associated with obesity
- **Findings**: HA, N/V, optic disc edema, transient visual obscurations, photopsias, VF defect, retrobulbar pain, pulsatile tinnitus, +/- vision loss, RAPD, diplopia, *elevated CSF pressure with normal composition (dx of exclusion)*
- **Etiology**: endocrinopathy, COPD, dural sinus thrombosis, use of steroids, OCP's, vitamin A, tetracycline, nalidixic acid use, isotretinoin
- **Tx**:
  - No vision loss: Headache (HA) treatment, weight loss
  - Mild vision loss: acetazolamide (Diamox), furosemide, weight loss
  - Advanced vision/VF loss: optic nerve sheath decompression, serial lumbar puncture (LP's), lumbar-peritoneal shunt

normal

Optic Neuritis

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- Inflammation of optic nerve
- MC optic neuropathy in patients <45 y/o
- **Findings**: acute unilateral visual loss (may recover spontaneously)
  - ocular pain on eye movement
  - decreased color vision
  - Relative afferent pupillary defect (RAPD)
  - visual field (VF) defect
  - *optic nerve appears swollen* ***or*** *normal*
  - Uhthoff's phenomenon: worsening of symptoms with heat or exercise
  - Phosphenes: flashes of light induced by eye movements or sound
- DDx: idiopathic, MS, syphilis, sarcoidosis, Lyme disease, Wegener's granulomatosis, SLE, Devic's disease
- *30% develop MS within 4 years; 38% develop MS within 10 years*

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Ocular Disease

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- May be isolated to 1 or both eyes
- May be related to systemic conditions
  - Infectious
  - Autoimmune
  - Neoplastic
  - Inflammatory
  - Traumatic

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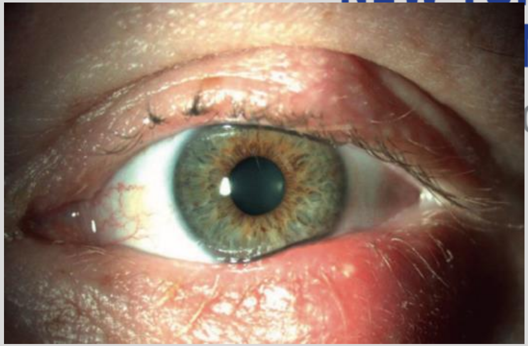
The Red Eye

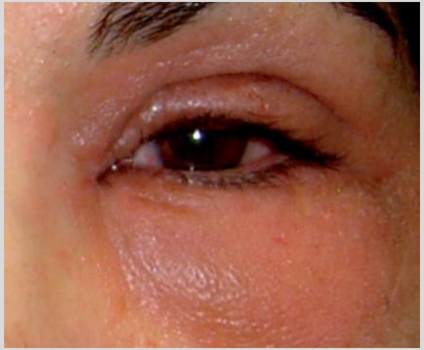
The Red Eyelid(s)

- Blepharitis
- Stye (external hordeolum)
- Chalazion (internal hordeolum)
- Preseptal Cellulitis
- Eyelid Abscess
- Herpes Simplex Dermatitis
- Herpes Zoster Ophthalmicus
- Orbital Cellulitis
- Trauma



← **Stye**

Chalazion→



← **Preseptal Cellulitis**

The Red Eyelid(s)

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Herpes Zoster Ophthalmicus

Orbital Cellulitis

Herpes Zoster Ophthalmicus

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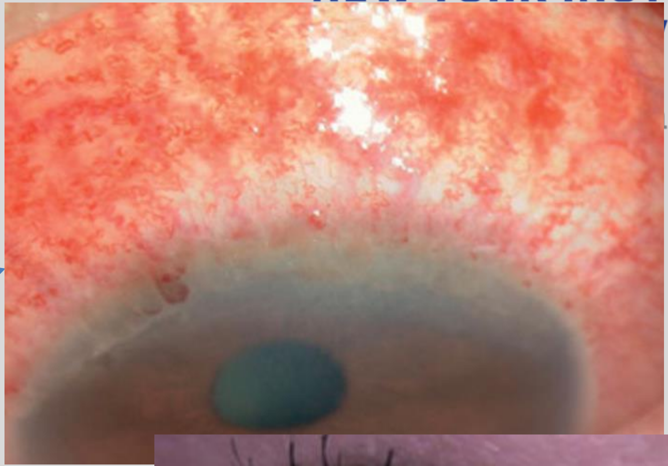
- **Unilateral** painful skin rash in a **dermatomal distribution** of the **trigeminal nerve**
  - Vesicles typically crust and heal within 2-6 weeks
  - **Hutchinson's sign**: skin lesions at tip, side, or root of the nose—strong predictor of ocular involvement (nasociliary nerve)
- Typically occurs in older adults (>60 y/o); can present at any age and occurs after reactivation of latent varicella-zoster virus (VZV) present within the sensory spinal or cerebral ganglia
    - HIV patients 15x more likely to have disease
  - VZV: dsDNA virus in herpes family
  - Shingles vaccine now recommended for patients >60y/o
  - Many cases have a prodromal period of fever, malaise, HA, eye pain prior to skin rash eruption
    - +/- eye pressure, tearing, eye redness, decreased VA, pain in distribution of trigeminal nerve
- **Management**: careful eye exam
  - Oral (PO) antiviral (acyclovir, famciclovir, or valcyclovir); immunosuppressed patients/those unable to tolerate PO may require IV tx
  - Topical antibiotic ointment to prevent superinfection
  - Topical steroids for keratitis/uveitis
  - PO/IV steroids for scleritis, retinitis, choroiditis, optic neuritis
  - Amitryptyline 25mg PO for neuropathic pain
- *Prognosis: variable and dependent on long term sequelae*

The Red Eye(ball)

- Conjunctivitis ("Pink Eye")
  - Allergic
  - Contact/Toxic
  - Inflammatory
  - Infectious
    - Viral
    - Bacterial
    - Fungal
    - Parasitic
- Subconjunctival Hemorrhage
- Episcleritis
  - Idiopathic
  - Autoimmune



Conjunctivitis Etiologies

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- **Viral**: MCC infectious conjunctivitis
    - Watery discharge, +- eyelid swelling
  - Adenovirus: highly contagious, incubation between 5-12 days
  - HSV-1: skin lesions may be present
- **Allergic**: observed most frequently in spring/summer
  - Lid swelling, diffuse conjunctival redness, chemosis
  - Tx: allergen avoidance, cold compresses, artificial tears, topical steroid, antihistamine(gtts/ PO)
- **Bacterial**: 2nd MCC, majority of cases in children
  - Direct contact, swimming pools, fomites
  - Usually lasts 7-10 days
  - Red eye, purulent/mucopurulent discharge, chemosis, decreases VA, +- eyelid swelling
  - Tx: hand hygiene!, symptomatic tx, cold compresses, artificial tears, (+/- betadine gtts?)
  - Tx: broad spectrum antbiotic gtts

- **Acute/subacute**:
  - Neisseria gonorrhoeae: risk of corneal perforation
    - Tx: 1g IM Ceftriaxone, lavage
  - Neisseria meningitides
  - Streptococcus pneumoniae
  - Haemophilus influenza
  - *COVID19
- **Chronic:** Staph aureus, Moraxella lacunata
- **Chlamydia trachomatis**
  - Azithromycin 1000mg PO x 1
  - doxycycline 100mg PO BID x 7 days
  - tetracycline 250mg QID x 7 days
  - erythromycin 500mg QID x 7 days

The Red Eye(ball)

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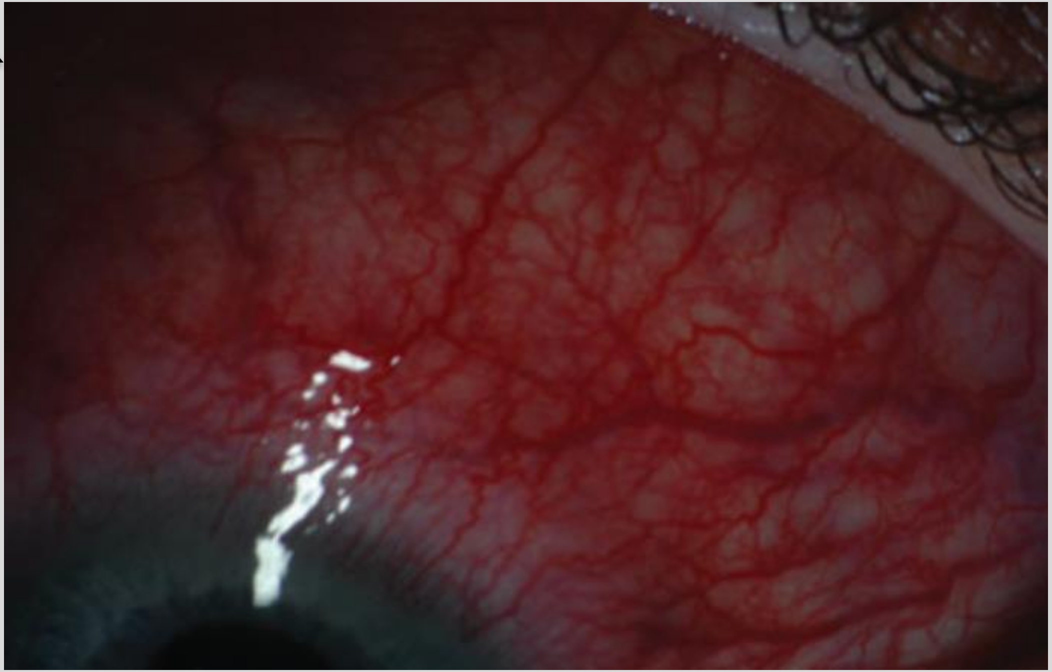
- **Scleritis**
  - Autoimmune
  - Infectious
- **Uveitis**
  - Iritis
  - Keratouveitis
  - Vitritis
  - Endophthalmitis



The Red Eye(ball)

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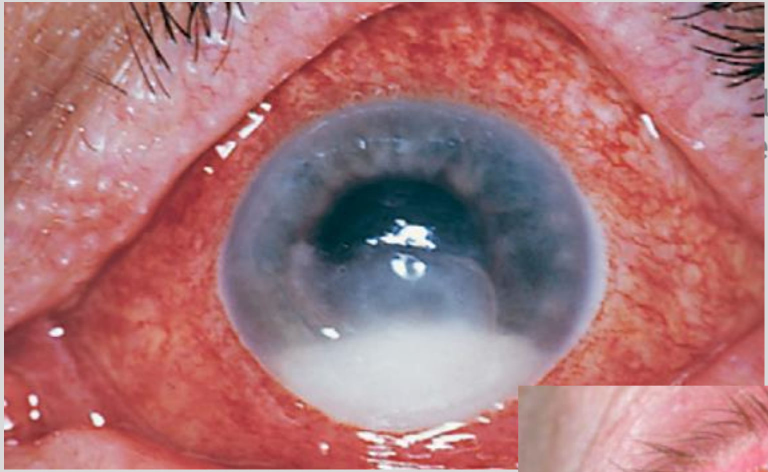
- Corneal Ulcer→
- Endophthalmitis
  - Post-surgical
  - Traumatic
  - Endogenous
- Glaucoma
  - Acute angle Closure Glaucoma
  - Open angle glaucoma



Endophthalmitis→

Acute Angle Closure Glaucoma

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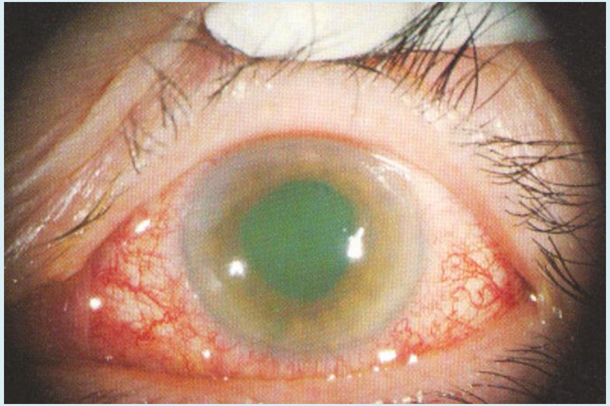
- Major cause of blindness worldwide
  - Higher prevalence in Inuit and Asian populations
  - Associated with increased age and **hyperopia** (**Farsightedness**)
- Secondary to closure of the anterior chamber drainage angle
- Typically **acute, dramatic attack** with increased intraocular pressure (IOP), nausea, vomiting, eye pain, eye redness, blurry vision, cloudy cornea, fixed, mid-dilated pupil
  - "I was sitting in a dark room when I suddenly had eye pain"
- Management: topical antihypertensive drops: goal is to lower eye pressure!
  - In severe cases, IV acetazolamide/mannitol may be necessary
  - **True emergency** – elevated IOP for a long period of time may cause permanent vision loss!
  - +/- topical saline/glucose for cloudy cornea
  - **Laser peripheral iridotomy: definitive treatment**



Differentiating Red Eyes

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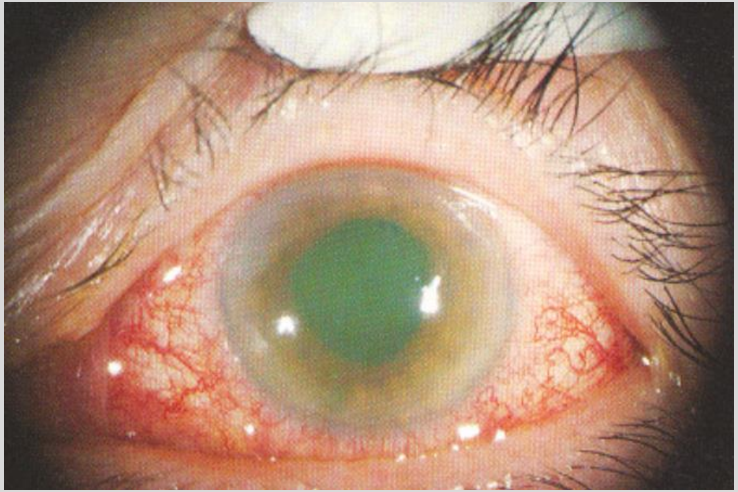


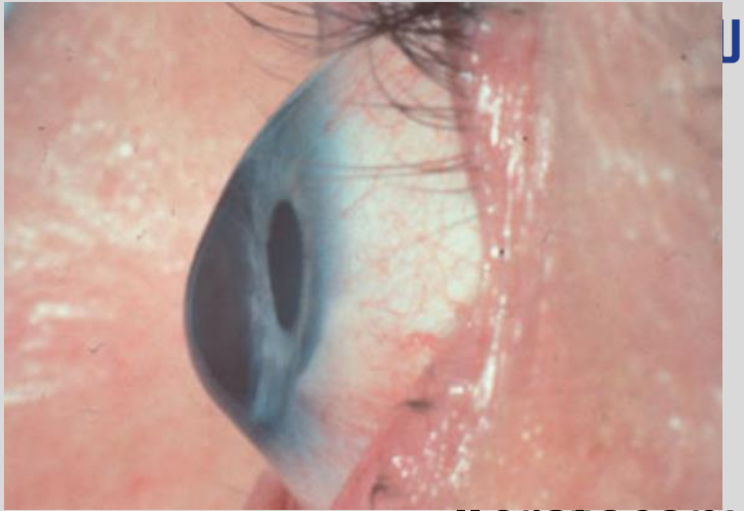
Is NOT =

(Angle Closure)

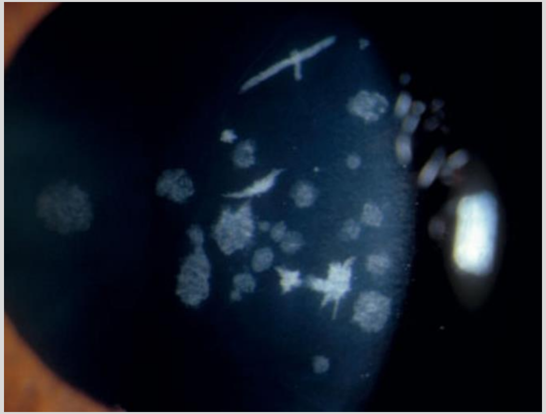
(Conjunctivitis)

Decreased Vision

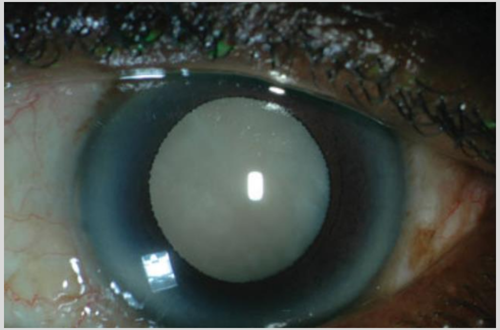
- Refractive error
  - Need for glasses/contact lenses
  - Myopia: near-sighted
  - Hyperopia: Far-sighted
- Degenerative conditions
  - Keratoconus
  - Corneal dystrophies
- Cataract



Keratoconus



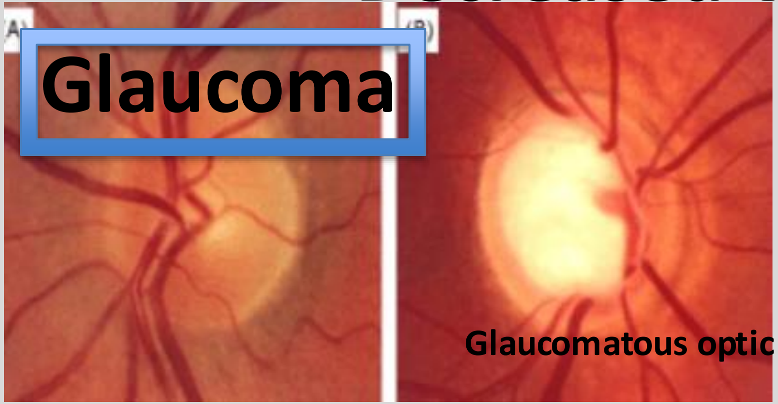
Granular Corneal dystrophy



Cataract

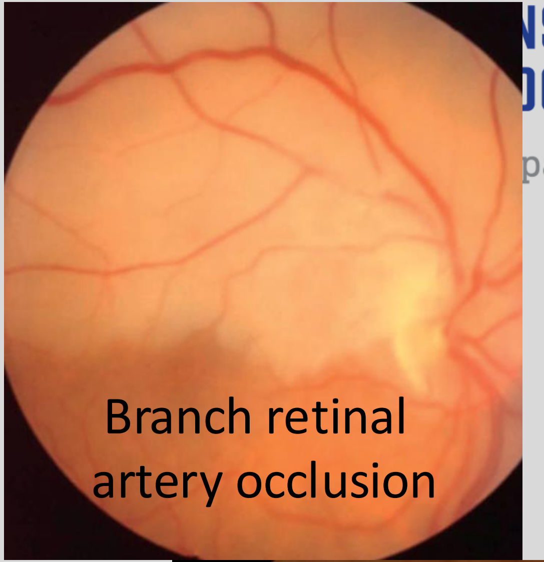
Decreased Vision

Glaucoma



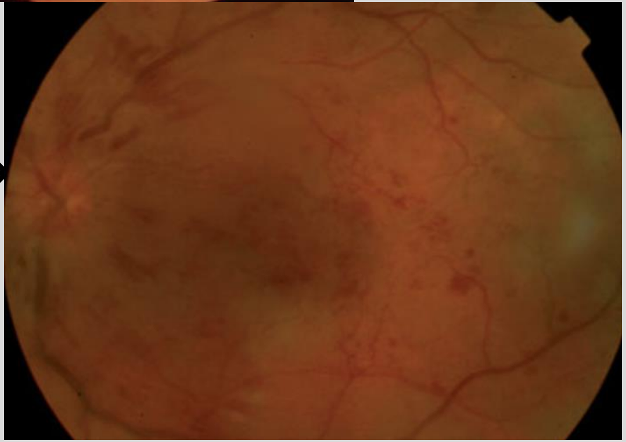
Glaucomatous optic neuropathy

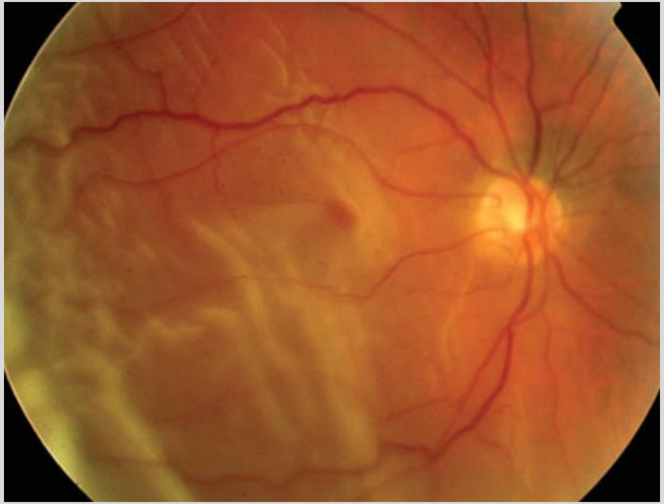
Retinal Vascular Occlusions



Branch retinal artery occlusion

Central retinal vein occlusion →

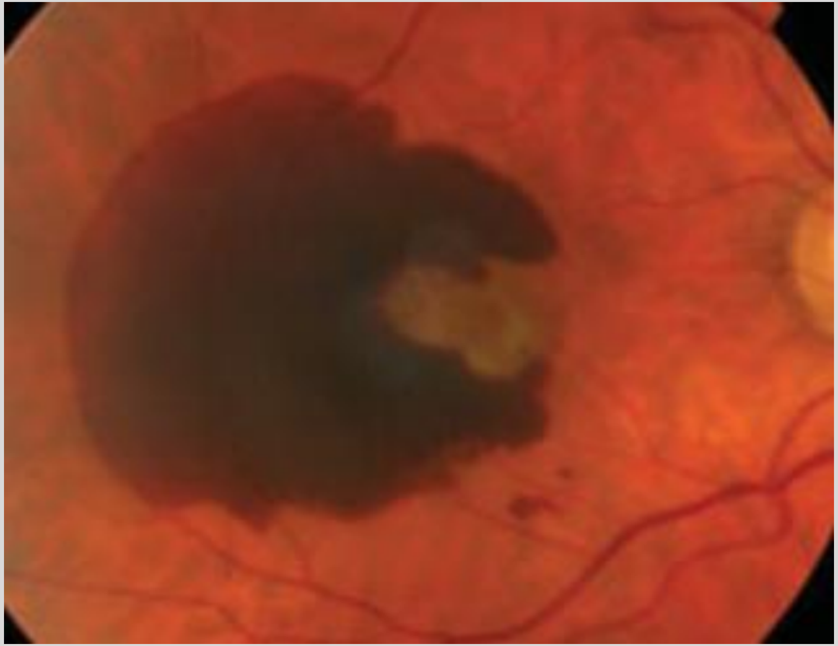




← Retinal Detachment

Decreased Vision

Dry AMD

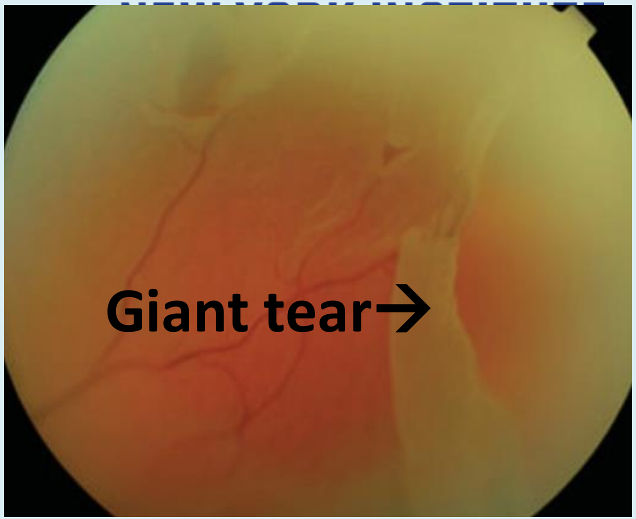


(Neovascular) Wet AMD

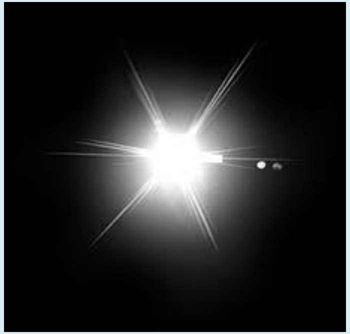
Age-Related Macular Degeneration (AMD)

Retinal Detachment

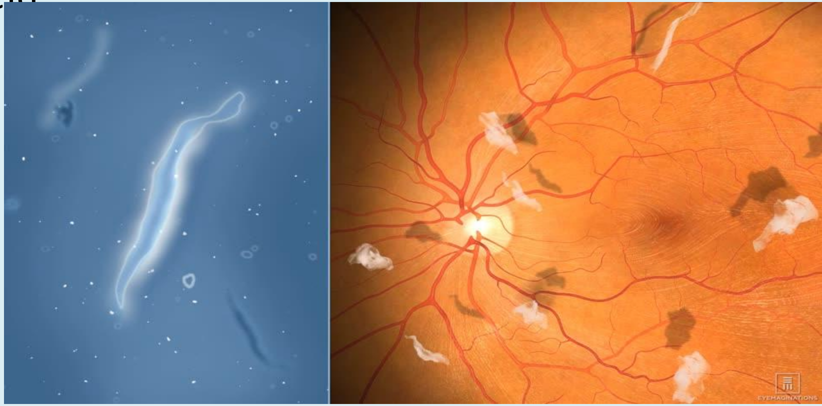
- Patient complaint: **flashes/floaters**
  - **Pertinent ocular history**: high myopia, recent eye surgery/trauma, family hx of retinal detachment, severe diabetic retinopathy, vitreous traction, intraocular tumor
  - **Flash**: camera flash/lightning (not sparklers…)
  - **Floaters**: multiple black spots
    - Curtain hanging down over/blocking vision/visual field
- Management: laser +/- vitrectomy surgery



"Flash"

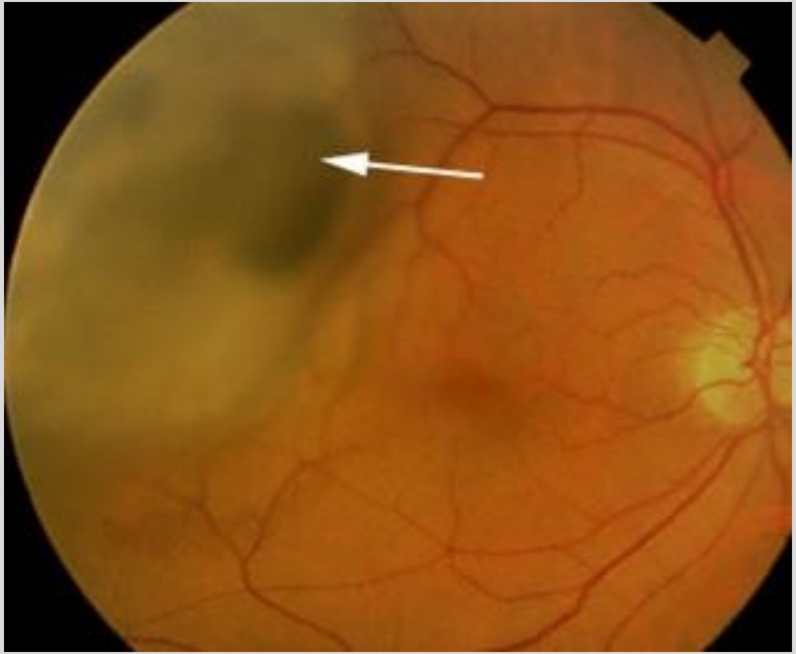


Floaters



Retinal Detachment due to Intraocular Mass

Uveal Melanoma: can be primary or metastatic



Leukocoria ("White Pupil")

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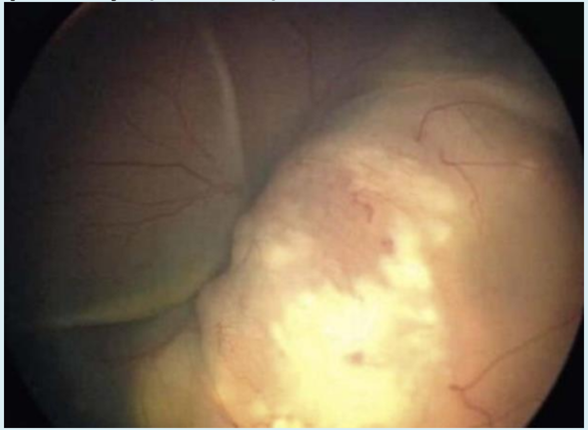
- **Retinoblastoma***
- Disorders of fetal vasculature: Coats disease, Persistent fetal vasculature (PFV), ocular toxocariasis, familial exudative vitreoretinopathy (FEVR), retinopathy of prematurity (ROP)
- Astrocytic hamartoma
- Vitreous hemorrhage, longstanding retinal detachment
- Endophthalmitis
- Coloboma



Retinoblastoma

Red reflex **White reflex**

Vision Loss

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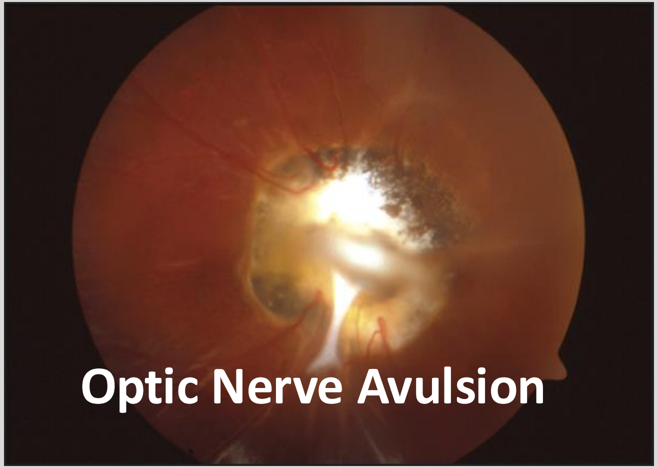
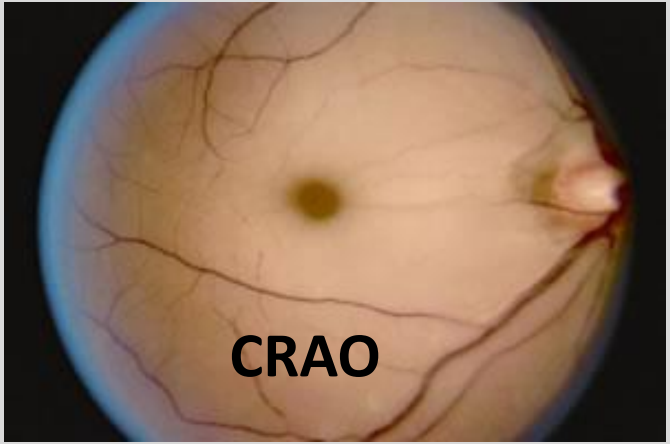
- Traumatic Optic Neuropathy
- Anterior ischemic optic neuropathy
  - Arteritic (*inflammatory*): Temporal Arteritis
  - Non-arteritic
- Posterior ischemic optic neuropathy
- Central retinal artery occlusion (CRAO)
- Central retinal vein occlusion (CRVO)
- Vitreous Hemorrhage





Acute Eye Pain

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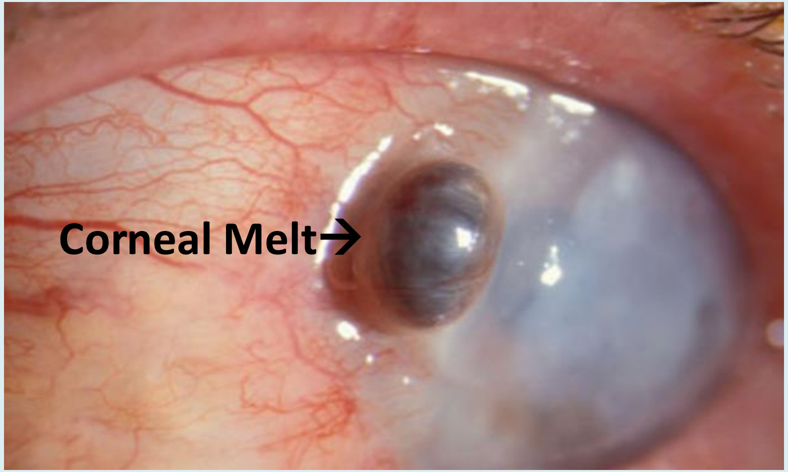
- Corneal abrasion/Ulcer
- Dry Eye
- Acute angle glaucoma
- Iritis
- Trauma

Corneal Abrasion

Corneal Abrasion

- Acute intense eye pain
- Usually h/o trauma or contact lens wear
  - *Recurrent erosion syndrome*: corneal epithelium irregular – easily abraded, therefore recurrent abrasions with minimal trauma (may even occur upon eye opening in the morning!)
- Management: antibiotic drops/ointment
  - +/- placement of bandage contact lens for large abrasions
  - Usually heals within a few days
  - DO NOT DISPENSE ANESTHETIC EYE DROPS→



Chronic Eye Pain

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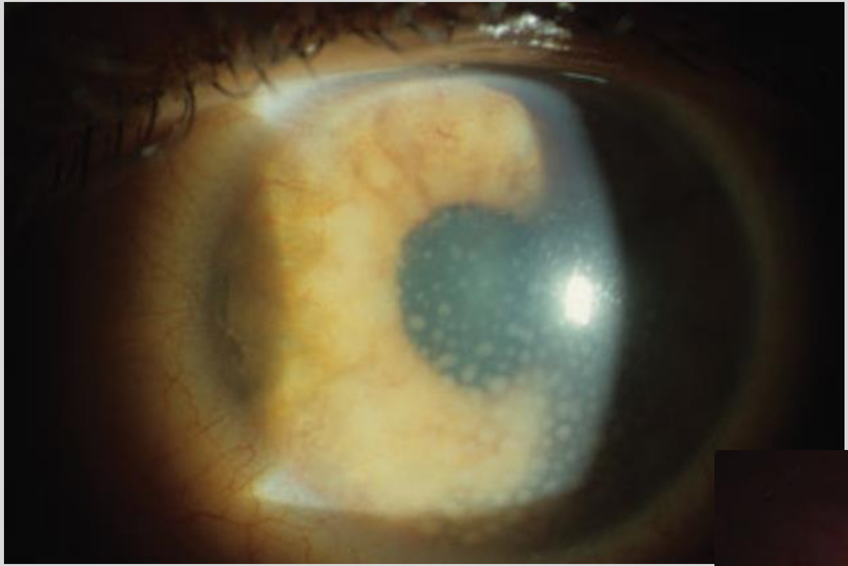
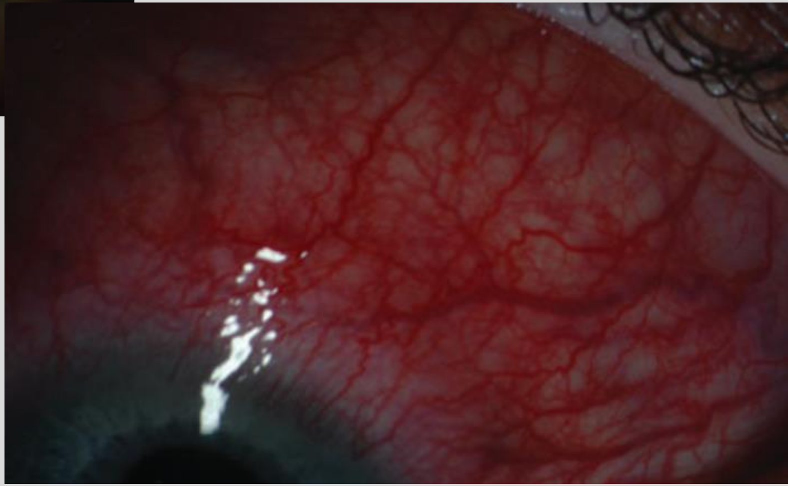
- Dry eye
- Chronic Uveitis
- Migraine
- Sinusitis
- Scleritis
- Episcleritis



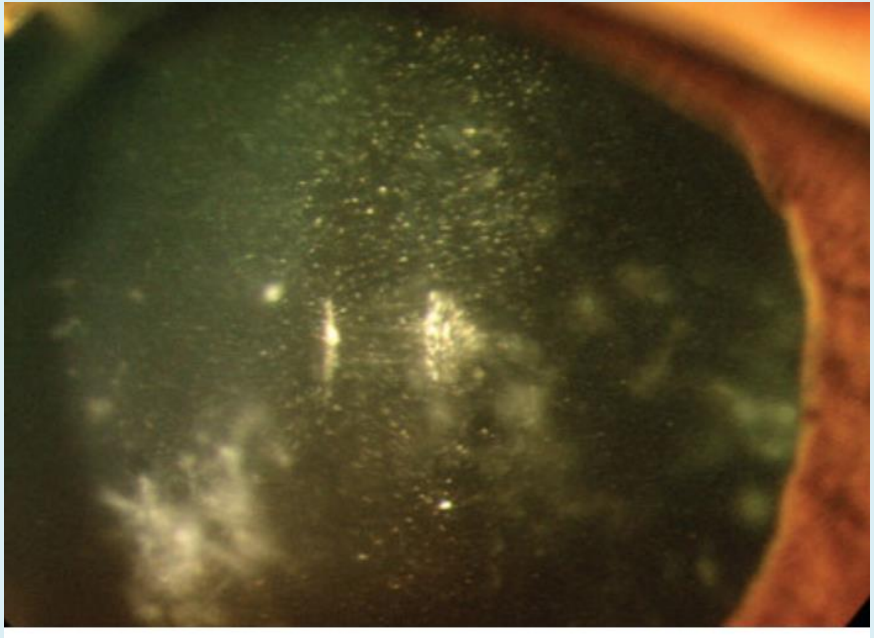
← **Chronic Uveitis with keratic precipitates**



Diffuse Scleritis→

Uveitis

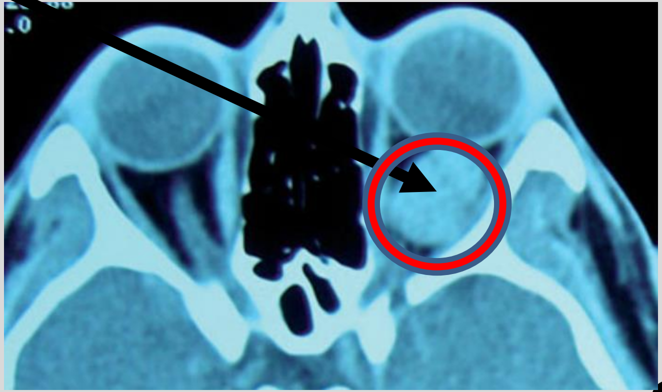
- Uvea: Iris, ciliary body, choroid
- Uveitis: inflammation of any/combination of these structures
  - Unilateral or Bilateral
- Etiology
  - Idiopathic – may only occur once!
  - Infectious
    - **Viral**: herpetic, HIV, CMV, TB, COVID-19
    - **Bacterial**: Syphilis, post-surgical infection
  - Autoimmune
    - Rheumatoid arthritis, lupus, Crohn's disease, Bechet's disease, Ankylosing spondylitis, Sarcoidosis
- Workup:
  - Patient history
  - Labs to check for infectious/systemic conditions that predispose to uveitis
  - Comprehensive ophthalmic exam
  - +/- referral to rheumatologist/infectious disease
- Treatment:
  - **Steroids**—topical, subconjunctival, periocular, intraocular, oral

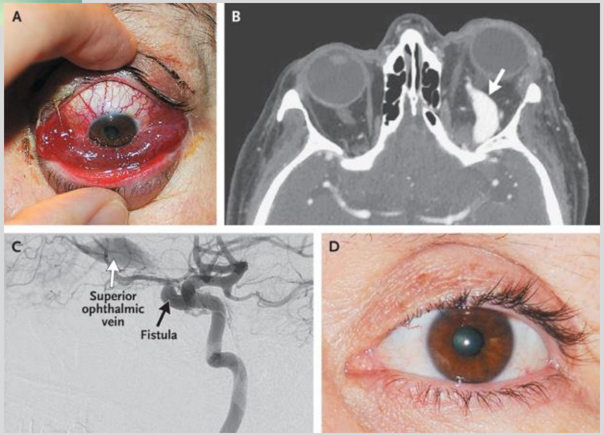


- **Common symptoms**: **dull achy pain, photophobia, blurry vision, floaters**
- **Common signs**: **poorly reactive pupil, conjunctival injection, keratic precipitates (KP's), white blood cells in the anterior (iritis) and/or posterior segment (*vitritis*)**

Diplopia

- Strabismus
- Thyroid Eye Disease
  - Proptosis
- Retroorbital Mass
- Ateriovenous malformation (AVM)
- Carotid-cavernous fistula





Carotid-cavernous fistula

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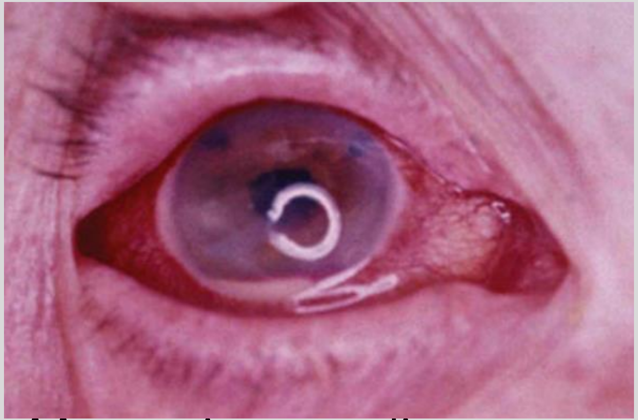
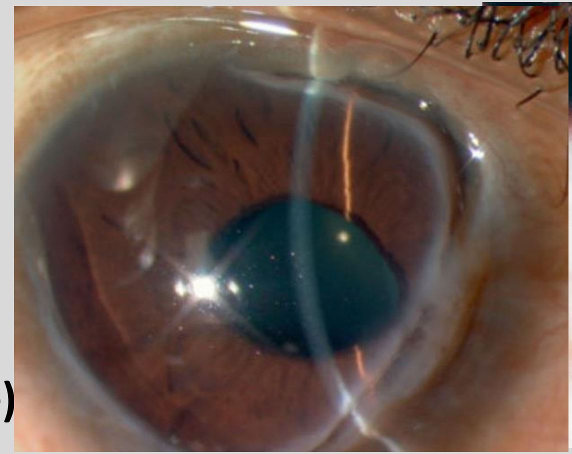
Systemic Disease with Ocular Findings

- Rheumatoid arthritis
- Lupus
- Sjogrens syndrome
- Crohn's Disease/IBD
- Wilson's disease
- Juvenile rheumatoid arthritis (JRA)
- Hepatitis C
- Syphilis



Uveitis: autoimmune disease

Corneal Thinning & vascularization (autoimmune disease)



Sunflower Cataract (Wilson's Disease)

Systemic Disease with Ocular Findings

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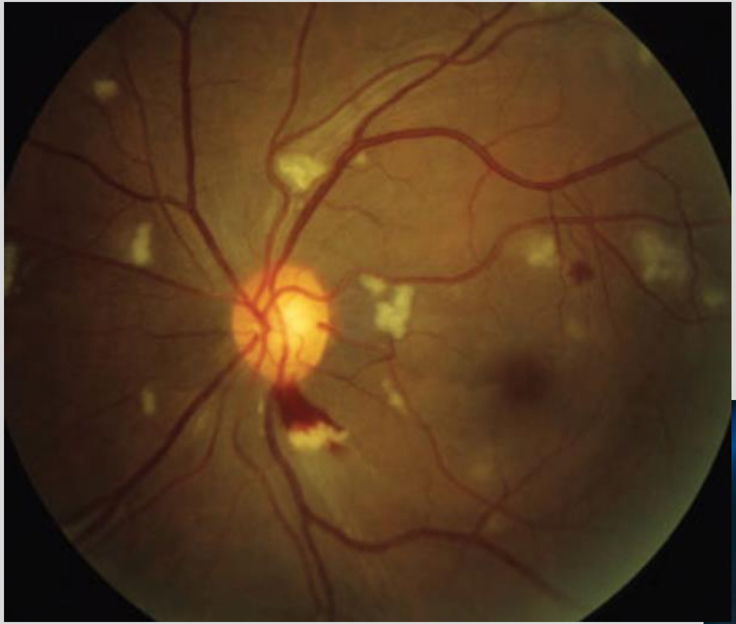
- Sarcoidosis
- TB
- Sepsis
- HIV
- Diabetes
- Hypertension
- Leukemia
- Melanoma



← **HIV Retinopathy**

Tuberculosis Retinitis→

Diabetic Retinopathy

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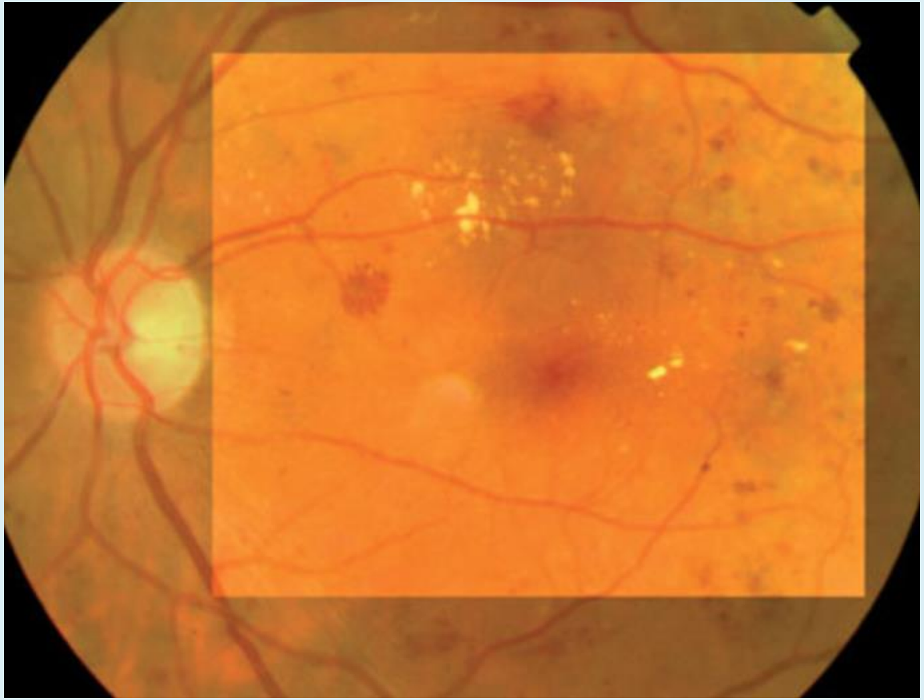
- Affects up of 80% of patient with Diabetes (I or II) >20yrs
  - Leading cause of blindness in patients 20-64 y/o
- Signs/Sx: blurry vision, distorted vision, floaters
  - Microaneurysms, retinal ischemia, macular edema, vitreous hemorrhage, retinal detachment (*a vasculopathy!)*
- Tx: GOAL IS TO reduce vision loss!
  - Laser surgery
  - Intravitreal injections: steroid, anti-Vascular endothelial growth factor (VEGF) agents
  - Vitrectomy surgery
  - <u>GLYCEMIC CONTROL</u>!



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Conclusions

- There are many ocular diseases and systemic conditions that effect the eye
- A good history and physical is key
- Always check vision, pressure (if possible) and refer when necessary!

Core References:

- Allen, Richard C. Basic Ophthalmology: Essentials for Medical Students. American Academy of Ophthalmology, 2016
- Ophthobook.com
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Lecture and Lab Feedback Form:

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